CS-TTD: Triplet Transformer for Compressive Hyperspectral Target Detection

Qingyao Yang, Xiaoqin Wang, Long Chen, Yumei Zhou, and Shushan Qiao, *Member, IEEE*

***Abstract*— Detecting targets in hyperspectral image under compressive sensing (CS) is crucial for real-time applications in front ends. However, applying the existing deep learning (DL) methods directly to compressed hyperspectral image has been challenging, including the uncertainty introduced by the random sensing matrix in the CS acquisition model, and the ineffectuality of traditional similarity metrics. In this article, we present a solution to these challenges by introducing a CS-based triplet transformer detector (CS-TTD). Our approach involves data augmentation using the restricted distribution property (RDP) to generate samples for a balanced number of positive samples, followed by the Siamese network with a triplet transformer to transition spectral vectors from the compressed domain to an embedding space. In addition, we introduce a combined convolution network (CCN) classification module to replace traditional metrics for obtaining classification results. To improve the training and make full use of dataset labels, we also present a two-stage training approach, known as intercategory separation and intracategory aggregation (ISIA), combined with hard-negative mining (HNM) and semi-HNM. Besides, to estimate the minimal number of compressed bands for sufficient detection performance, we introduce a kernel distribution estimation-based sequential iteration (KDE-SI). Our method achieves hyperspectral target detection (HTD) without reconstruction, and experimental results demonstrate that its performance at 0.16 compression ratio (CR) is comparative to the existing methods in the original domain. The code for this work is available at https://github.com/nicyyyy/CS-TTD.git.**

***Index Terms*— Compressive sensing (CS), hyperspectral, restricted distribution property (RDP), Siamese network, target detect, transformer.**

I. Introduction

HYPERSPECTRAL imaging, an advanced optical sensing technology, holds great application potential. It captures spectral information across numerous narrow bands, facilitating substance differentiation. It is applied extensively in agriculture monitoring [1], [2], [3], [4], disaster warning [5], [6], medical imaging [7], [8], and military reconnaissance [9], [10], [11]; hyperspectral imaging is vital for target detection, utilizing prior spectral information to identify characteristic spectra [12], [13].

Manuscript received 29 June 2024; accepted 28 July 2024. Date of publication 31 July 2024; date of current version 9 October 2024. *(Corresponding author: Xiaoqin Wang.)*

The authors are with the Institute of Microelectronics, Chinese Academy of Sciences, Chaoyang, Beijing 100029, China, and also with the School of Integrated Circuits, University of Chinese Academy of Sciences, Huairou, Beijing 101408, China (e-mail: yangqingyao@ime.ac.cn; wangxiaoqin@ime.ac.cn; chenlong@ime.ac.cn; ymzhou@ime.ac.cn; qiaoshushan@ime.ac.cn).

Digital Object Identifier 10.1109/TGRS.2024.3436084

Compressive sensing (CS) is an advanced method that has found extensive applications in snapshot hyperspectral imaging technology [14]. This technique reduces sampling frequency through incoherent sampling, offering advantages, such as low-memory usage, low-power consumption, low bandwidth, and cost-effectiveness [48]. However, the reconstruction of CS is time-consuming and resource-intensive, making it impractical for real-time detection and target tracking on front-end platforms [15]. Consequently, there has been significant research on data processing techniques under CS, which benefits real-time tasks and low-power front-end platforms systems [16], [17], [18]. This article includes an analysis of traditional hyperspectral target detection (HTD) methods and proposes HTD methods for the compressive hyperspectral image.

A. Traditional HTD

Previous HTD is based on original hyperspectral. Classical HTD methods match prior spectra with the spectra to be detected, emphasizing differences for background and target separation. Orthogonal subspace projection (OSP) [19] projects hyperspectral images, using differences between target and background subspaces for HTD. Adaptive coherent estimator (ACE) [20], based on an unstructured background model, solves the maximum likelihood ratio. Spectral matched filter (SMF) algorithm, using a linear model assumption, matches and detects using known spectral features and background clutter covariance matrices. Constrained energy minimization (CEM) [21] applies constraints based on finite impulse response filters for both target and background, minimizing the filter output response for detection. Building upon this, improved algorithms emerged, such as h-CEM [22] cascading multiple layers of detectors and ensemble-based E-CEM [23]. HTD methods based on hyperspectral representation features formulate an optimization problem to express targets and backgrounds as combinations of atomic dictionaries, separating them based on differences [24], [25]. For instance, composite sparse and collaborative representation (CSCR) [26] detector utilizes target and background libraries for detection. Sparse-based target detection (STD) [27] reconstructs hyperspectral images using an atomic dictionary, extracting targets from reconstruction residuals.

In recent years, HTD methods based on deep learning (DL) and Siamese networks have also been explored. Siamese networks, with their two identical subnetworks and shared

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

weights, exhibit strong performance to provide robust nonlinear feature representations for spectral samples [28], [29], [30]. LRS-STD [31], in prior research, introduced a target low-rank sparse detector using a sparse autoencoder and a Siamese fully connected network based on assumptions of background low-rank and target sparsity. STTD [32] utilized high-purity pixel samples mixed with prior target spectra, training a Siamese transformer with similarity measures. TSTTD [33], an improved version, introduced a triplet transformer trained on sample pairs using a radiative transfer model and proposed the intercategory separation and intracategory aggregation (ISIA) loss to enhance target-background separability. MLSN [34] utilizes deep residual convolution for spectral vector embedding into Euclidean space, while meta-training with known labeled source data on a three-channel Siamese network to learn spectral intraclass similarity and interclass dissimilarity.

The existing Siamese networks still face various challenges in compressive hyperspectral.

Problem 1: The utilization of DL-based HTD methods frequently necessitates data augmentation to create spectral vectors and a class-balanced training dataset. These methods often depend on generative models, such as AE [35], [36] and GAN [37], [38], to extract spectral features and generate synthetic spectral samples. Nevertheless, they do not accommodate the uncertainties arising from the sensing matrix and sparse matrix in CS.

Problem 2: The existing Siamese networks commonly utilize cosine similarity in metric learning, a method that may not effectively measure similarity in the compressed domain. According to the restricted cosine property (RCP), the angle between spectral vectors under CS undergoes a linear transformation with the following properties [41], [42]. α is the angle between vectors in the original domain, θ is the angle between vectors in the CS domain, and δ_k is the restricted isometry constant (RIC). This implies that under CS, the angle between vectors of the same class may change

$$\frac{(1-\delta_k)}{(1+\delta_k)}\cos\theta \le \cos\alpha \le \frac{(1+\delta_k)}{(1-\delta_k)}\cos\theta. \tag{1}$$

Moreover, cosine similarity, based on the angle between vectors, is highly sensitive to vector direction but does not effectively distinguish feature vectors of different lengths along the same direction.

Problem 3: How to estimate the minimum number of compressed bands required for HTD methods. The performance of algorithms for compressed hyperspectral processing is affected by the number of compressed bands, and there exist a minimum number of bands required to maintain high detection performance [43], [44]. Identifying this number helps to reduce the computation.

For problem 1, a data augmentation method based on restricted distribution property (RDP) is proposed. In our previous work [40], we introduced the concept of RDP within compressed sensing (CS), which preserves the distribution of spectral random vectors during the CS process. The original domain's target spectral vector is perceived as a random vector due to spectral variability, and its distribution in CS adheres to a Gaussian distribution linked to the variance of random vectors in the original domain. The proposed augmentation method entails the generation of a random vector based on this distribution to serve as the target spectral samples. This approach is straightforward and can be seamlessly integrated into the training process without necessitating any additional structure.

For problem 2, a novel network architecture known as the combined convolution network (CCN) is proposed to discern positive samples from the negatives. Within the CCN framework, the spectral vector slated for classification is amalgamated with an anchor. The CCN model adeptly learns nonlinear relationships between features and effectively utilizes sample labels, making it more efficient than using only metric learning approaches.

As the backbone of Siamese network, the triplet transformer is constructed to extract global features with long-range dependencies from the compressed spectral vectors. To jointly train triplet transformer and CCN, we adopt two-stage ISIA, which is a semisupervised approach that combines hard-negative mining (HNM) triplet loss [45], [46] with binary cross-entropy (BCE) loss.

For problem 3, a kernel distribution estimation-based sequential iteration (KDE-SI) method is proposed to estimate the minimum number of compressed bands for the proposed method. According to RDP, the compressed spectral vector follows a Gaussian distribution. The RDP-based data augmentation mentioned above generates positive samples based on this distribution. When the number of bands is too low, the original positive samples fail to exhibit this distribution, causing a mismatch with the generated positive samples. Thus, the problem becomes estimating the minimum number of bands needed to represent the Gaussian distribution. Kernel density estimation (KDE) is a method for estimating the probability function, and this article introduces a sequential iterative algorithm based on KDE to find the minimum number of bands.

In summary, this article introduces a novel approach named CS-based triplet transformer detector (CS-TTD), with RDP-based data augmentation and CCN for spectral classification. This achieves HTD in CS without reconstruction. A KDE-SI is also proposed to estimate the minimal CS bands number. The contributions of this article are as follows.

1) The proposal RDP-based data augmentation, which samples the target spectral distribution under CS, can be integrated into training process.
2) The introduction of a CCN structure for classification prediction, combined with triplet transformer as the backbone of Siamese network.
3) A two-stage ISIA semisupervised training method is also adopted to jointly train the networks.
4) The development of KDE-SI method to estimate the minimum number of compressed bands for the proposed method.

II. Relative Works

CS has become a vital technology in information theory, communication, and signal processing. By employing a nonadaptive strategy, CS efficiently recovers signals of interest under the Nyquist sampling rate, reducing sampling speed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

and data storage needs. This revolutionary approach directly acquires compressed information, significantly impacting communication and signal processing by achieving objectives while lowering hardware costs and computational requirements.

A hyperspectral image $Y \in \mathbb{R}^{H\times W\times L}$ can be decomposed into spatial dimensions of size $H\times W$ and a spectral dimension of length L. The sparse sampling of the vector's spectrum from the original L-bands to M spectral bands, where $M \ll L$, can be expressed as follows:

$$y = \Phi\Psi r + n. \tag{2}$$

Here, $y \in \mathbb{R}^M$ represents the sparsely sampled spectral vector, $r \in \mathbb{R}^L$ is the spectral vector in original domain, $\Phi \in \mathbb{R}^{M\times L}$ is the sensing matrix, $\Psi \in \mathbb{R}^{L\times L}$ is the sparse matrix, and n denotes noise.

In CS theory, it is required that the original signal r exhibits sparsity, or it can be transformed into a sparse signal using a sparse transform matrix Ψ. Moreover, the sensing matrix Φ should be as incoherent as possible with the sparse matrix Ψ. The coherence is defined as (3), where Φ_j and Ψ_j represent the columns of Φ and Ψ

$$1 \le \mu(\Phi, \Psi) = \sqrt{L} \quad \max_{1\le j,i\le L} |\langle \Phi_j, \Psi_i\rangle| \le \sqrt{L}. \tag{3}$$

Given a sparse matrix, it is necessary to select a sampling matrix in a way that minimizes coherence while satisfying the restricted isometry property (RIP) [47]. RIP can be expressed as the following inequality:

$$(1-\delta_k)\|r\|_2^2 \le \|\Phi r\|_2^2 \le (1+\delta_k)\|r\|_2^2. \tag{4}$$

Here, δ_k is the RIC in the range of 0–1.

The significance of RIP can be understood as follows: for any k-sparse vectors, after transforming by the sensing matrix Φ, their Euclidean distances are preserved. It ensures the precise recovery of compressed signal. Random sensing matrices have been previously proven to satisfy the RIP and incoherent with any sparse matrix [47].

On that basis, Porta et al. [48] propose a universal sensed model by constructing a Gaussian distributed sensing matrix Φ_G, where each element Φ_{ij} satisfies $\Phi_{ij} \sim \mathcal{N}(0, 1/L)$. In this case, the sparse matrix Ψ does not need to be known during signal acquisition, only during signal reconstruction. CS under the universal sensed model can be simplified to the following formula:

$$y = \Phi_G r + n. \tag{5}$$

This indicates that, under the universal sensed model, algorithm can proceed without the need to precisely know the sparse matrix. Moreover, all the information is retained in the CS signal, as confirmed in [47].

Building upon the foundation of the universal sensed model, our prior work introduced a preserved property called RDP under CS [40]. Due to various disturbances, the target spectrum in the original domain exhibits spectral variability and can be represented as a spectral random vector. Each band $r_t(l)$ is an independent random variable

$$r_t = [r_t(1), r_t(2), \ldots, r_t(L)]^{\mathrm{T}}. \tag{6}$$

According to RDP, each element of the target spectrum y_t under CS follows a Gaussian distribution:

$$y_t(i) \sim \mathcal{N}\left(0, B_L^2\right) \tag{7}$$

where B_L^2 is the RDP coefficient, which can be calculated based on the mean $\mu_t(l)$ and variance $\sigma_t^2(l)$ of each band in the original domain. The derivation is attached to the [40]

$$B_L^2 = \frac{1}{M}\sum_{l=1}^{L}\left(\mu_t^2(l) + \sigma_t^2(l)\right). \tag{8}$$

III. Proposed Method

The CS-TTD proposed in this article consists of a triplet transformer for feature extraction and a CCN for classification. During training, the target's distribution under CS is obtained via RDP, sampled as an anchor, and processed through shared-weight transformers and CCNs. In the detection phase, pixel spectra pass through the transformer and, along with the anchor, input to CCN, producing the detection map. The overall framework is shown in Fig. 1.

A. RDP-Based Data Augment for Triple

In the training of triplet transformer, the input data consist of positive, negative, and anchor, denoted as a triple $[y_+, y_-, y_a]$. These samples are input into three transformers with shared weights.

According to RDP, the distribution of the target spectrum in CS follows $\mathcal{N}(0, B_L^2)$ in (7). At the beginning of one training epoch, a random vector of length M that follows $N(0, 1)$ is generated. This vector is then transformed to follow $\mathcal{N}(0, B_L^2)$, as shown in (9). This sample is seen as the anchor y_a in the triple

$$y_a = B_L^2 \cdot \mathrm{random}(1, 1, M). \tag{9}$$

This data augmentation approach is outlined in Fig. 2. The anchor y_a randomly drawn from the RDP distribution can be interpreted as a sample that falls within the same category as the target. In hyperspectral data, the quantity of target samples is generally much lower than that of the background. This sampling technique effectively boosts the number of positive samples, thereby mitigating the problem of class imbalance.

B. Triplet Transformer

The triplet transformer consists of three weight sharing transformer encoders. Each transformer encoder includes MultiHead attention and feedforward neural network (FNN).

1) MultiHead Attention: This article employs triplet transformer to capture global dependencies in 1-D spectral samples, embedding them into feature vectors in a feature space. In the proposed method, three fully identical transformers with the same parameters are constructed, achieving distinct mappings of positive and negative samples in the embedded space through metric learning.

A simple embedding is applied to the input spectral vector, mapping $y \in \mathbb{R}^{1\times M}$ to $y^e \in \mathbb{R}^{N\times M_s}$. To be more specific, the 1-D spectral vector is evenly divided into N groups, each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

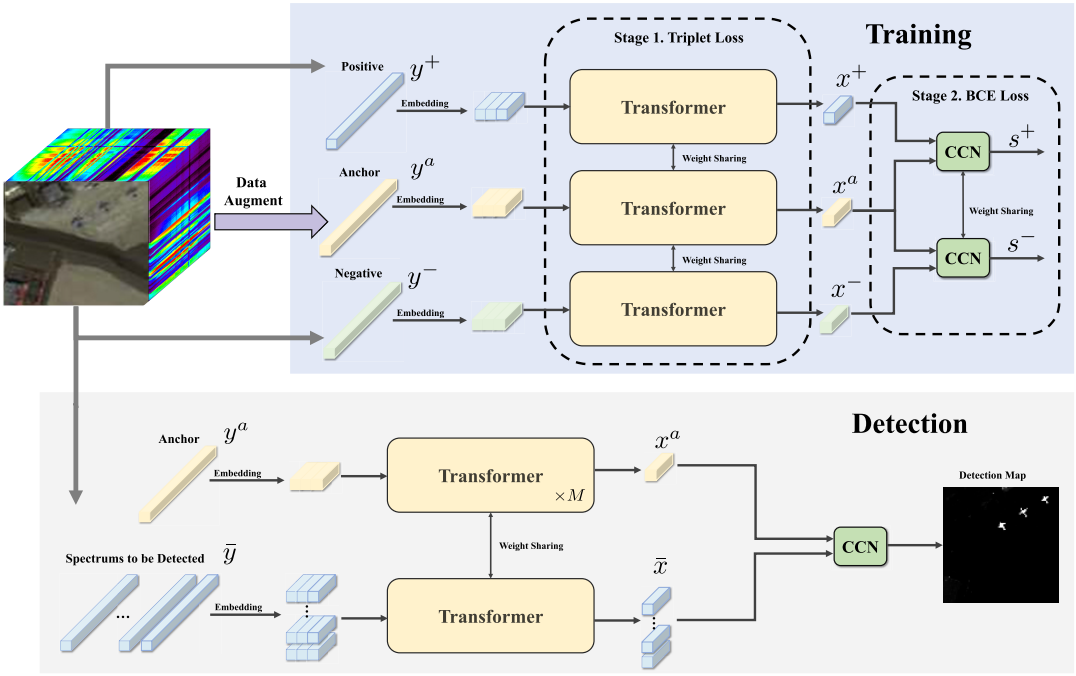


Fig. 1. Framework of the proposed method.

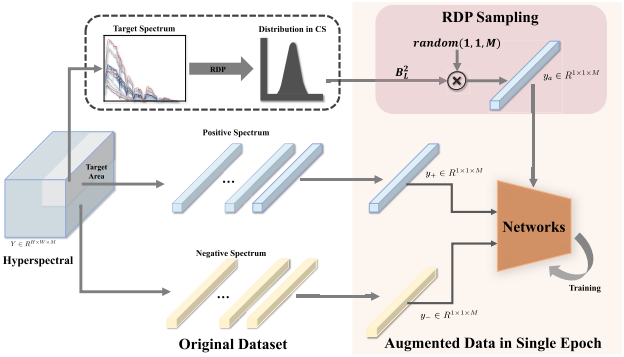


Fig. 2. RDP-based data augmentation.

of size M_s. Each row of the embedding matrix is denoted as $y_i^e \in \mathbb{R}^{1\times M_s}$.

The y^e is then fed into the multihead attention mechanism, a core component of the transformer. This mechanism applies multiple attention heads in parallel, allowing the model to learn feature representations in different subspaces simultaneously. The input is mapped into query matrix Q, key matrix K, and value matrix V through linear transformation matrices

$$\begin{aligned} Q_i &= y^e W_{Qi} \\ K_i &= y^e W_{Ki} \\ V_i &= y^e W_{Vi}. \end{aligned} \tag{10}$$

The matrices W_{Qi}, W_{Ki}, and W_{Vi} represent the weight matrices for the query, key, and value in the ith attention head, respectively. The attention distribution is computed using the Q, K, and V matrices

$$\text{head}_i(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^{\mathrm{T}}}{\sqrt{d_k}}\right) V_i. \tag{11}$$

Here, softmax function is used to normalize the attention distribution, and $(d_k)^{1/2}$ is employed to alleviate the gradient vanishing problem within each attention head, where d_k is the dimensionality of each key. Subsequently, the outputs from multiple attention heads are concatenated to compute the overall output

$$\text{MultiHead}\left(y^e\right) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h) W_O. \tag{12}$$

W_O is the output weight matrix, and h represents the number of attention heads in the multihead attention mechanism.

2) Feedforward Neural Network: The FNN consists of two layers of fully connected layers. The activation function for the first layer is ReLU, and the second layer does not use any activation function

$$X = \max(0, XW_1 + b_1)W_2 + b_2 \tag{13}$$

where X is the input to the FNN, and the input and output sizes are the same. W and b represent the parameters of the two linear layers.

3) Add and Norm: The "Add" operation is a residual connection commonly found in ResNet architectures. "Norm" refers to layer normalization, typically employed in RNN structures. Layer normalization adjusts the inputs of each neuron in a layer to have the same mean and variance, promoting faster convergence. In this article, "Add and Norm" operations are introduced in two instances: MultiHead attention and FNN.

The transformer network can deeply extract global dependency features from spectral vectors through cascaded layers. Finally, the outputs are summed along the columns to obtain the output vector $x \in R^{1\times M_s}$

$$x = \text{sum}(X, \text{dim} = 1). \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

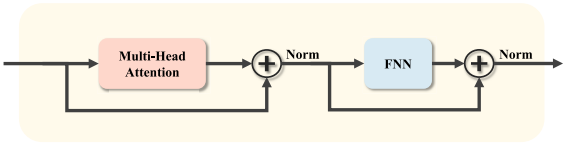


Fig. 3. Structure of transformer.

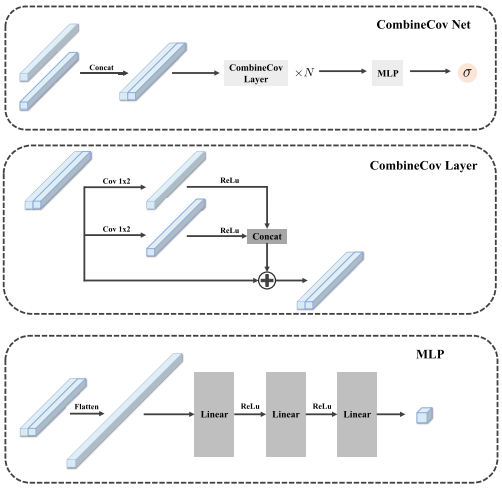


Fig. 4. Structure of CCN.

The transformer structure is depicted in Fig. 3.

C. CombineCovNet

In this section, the CCN for supervised learning is introduced, as illustrated in Fig. 4. The input to this section consists of two feature vectors after transformer embedding, $x_1, x_2 \in \mathbb{R}^{1\times M_s}$, which are concatenated to form a new 2-D matrix $x_c \in \mathbb{R}^{2\times M_s}$

$$x_c = \text{Concat}(x_1, x_2). \tag{15}$$

The x_c is fed into a series of cascaded CombineCov layers. Each layer consists of two convolutions of size 1×2, followed by ReLU activation, resulting in two vectors $t_1, t_2 \in R^{1\times M_s}$

$$\begin{aligned} t_1 &= \max(0, x_c * \omega_1) \\ t_2 &= \max(0, x_c * \omega_2). \end{aligned} \tag{16}$$

Here, ω_1 and ω_2 are convolutional kernels, and $*$ denotes the convolution operation. Concatenating t_1 and t_2 yields a matrix of the same size as the input, which is then added to the input x_c to obtain the output of this layer

$$x_L = x_c + \text{Concat}(t_1, t_2). \tag{17}$$

The multilayer perceptron (MLP) consists of three fully connected layers and two ReLU activation functions. Its input is the flattened x_L, and the parameters are denoted by θ

$$x_M = \text{MLP}(x_L, \theta). \tag{18}$$

Finally, the network outputs the probability that the feature vectors x_1 and x_2 belong to the same category using the sigmoid activation function

$$s = \sigma(x_M). \tag{19}$$

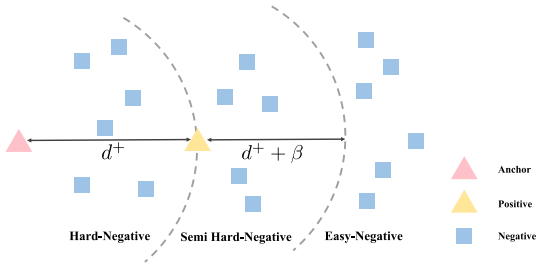


Fig. 5. Illustration of three negative.

In CCN, two input feature vectors come from the sample to be detected and the anchor, and they are aligned and combined. Two 1-D convolutions are applied to extract local information in different spaces. On the other hand, the residual connection ensures the retention of global features while preventing the gradients from vanishing when cascading multiple CombineCov layers. This enables the deep convolutional learning of the nonlinear relationship between the two feature vectors. Finally, a multilayer perceptron with a sigmoid activation function maps this nonlinear relationship to the probability that the samples belong to the same category. Training is performed using labels and BCE loss function.

D. Two-Stage ISIA

1) Transformer Training: The basic unit for training the triplet transformer is a triple, consisting of an anchor, a positive sample, and a negative sample. By embedding them into the feature space using triplet transformer, we obtain the feature vectors $[x^a, x^+, x^-]$. Let the distance from the anchor to the positive sample in the embedded space be denoted as d^+, and the distance from the anchor to the negative sample as d^-. The distance metric involves sampling the Euclidean distance between the feature vectors

$$\begin{aligned} d^+ &= \|\text{anchor} - \text{positive}\|_2 \\ d^- &= \|\text{anchor} - \text{negative}\|_2. \end{aligned} \tag{20}$$

The triplet loss function aims to minimize the distance between the anchor and the positive sample while maximizing the distance between the anchor and the negative sample, thereby enhancing the discriminative ability of the learned features. This strategy helps create a more robust representation in the feature space. For a single triple, the triplet loss is expressed as (21), where α is the margin, defining the minimum expected difference between positive and negative pairs

$$L_{\text{Triplet}} = \max\left(0, d^+ - d^- + \alpha\right). \tag{21}$$

In triplet loss, the selection of negative samples is crucial. Negative samples are typically categorized into three types: hard negative, semi-hard negative, and easy negative. Fig. 5 provides an intuitive representation of these three types of negatives in the feature space in relation to the anchor and positive samples.

Hard negative refers to negative samples that have a smaller embedding distance from the anchor, i.e., $d^- < d^+$. In this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

case, negatives are closer to the anchor compared with positives, making it challenging for the model to correctly classify them due to their smaller distance in the embedding space. During training, selecting such hard-negative samples helps the model learn better boundaries, enhancing the discriminative nature of the embeddings.

Semi-hard negative refers to negative samples with embedding distances to the anchor slightly farther than the distances from positives to the anchor, but the difference in distances does not reach a specified margin β. This condition may lead to misclassification to some extent.

Easy negative refers to negative samples with embedding distances from the anchor that are relatively far. Choosing such negative samples during training may lead the model to converge earlier, but it might not fully exploit the differences in the sample space, resulting in less discriminative embeddings.

A challenge in training with triplet loss is the selection of triplets. Simply choosing random triplets may result in slow convergence and poor embedding quality. This article addresses this challenge by employing HNM to select triplets.

In the initial training stage, an anchor is randomly sampled and fixed for that training round. The distances d_j^- from all positives to the anchor are computed. For a positive sample with distance d^+ to the anchor, the hard-negative samples satisfying $d_j^- < d^+$ are selected for backpropagation, with the total number of N.

Then, the HNM triplet can be expressed as (22), penalizing case that the distance from positive samples to the anchor exceeds that from negative samples to the anchor

$$L_{\text{Triplet}} = \frac{1}{N}\sum_{j=1}^{N}\max\left(0, d^+ - d_j^- + \alpha\right). \tag{22}$$

2) CCN Training: The embedded triple vectors $[x^a, x^+, x^-]$ from the triplet transformer are split into $[x^a, x^+]$ and $[x^a, x^-]$ and fed into two CCN modules with shared weights. The goal is for the network to classify $[x^a, x^+]$ with a result close to 1 and $[x^a, x^-]$ with a result close to 0.

After completing the transformer training stage, the network has acquired the ability to separate positive and negative samples in the embedding space. The majority of negatives are encoded in a region far from the anchor, resulting in a significant reduction in the number of existing hard negatives, which is insufficient for adequate training of CCN. To address this, the criteria for selecting negatives are appropriately relaxed, and M semi-hard negatives are chosen with $d_j^- + \beta < d^+$, forming M $[x^a, x^+]$ and $[x^a, x^-]$ pairs. After passing through CCN, classification probabilities s^+ and s_j^- are obtained. With labels $s_t^+ = 1$ and $s_t^- = 0$, a BCE loss is constructed for backward propagation to update CCN parameters

$$L_{\text{BCE}} = -\frac{1}{M}\sum_{j=1}^{M}\left[s_t^+ \cdot \log\left(s^+\right) + \left(1 - s_t^-\right) \cdot \log\left(1 - s_j^-\right)\right]. \tag{23}$$

In summary, the training of CCN and transformer is mutually exclusive, with one set of parameters fixed while training the other. The first stage primarily focuses on developing the partitioning ability of the embedding space feature vectors based on an unsupervised learning. The second stage is dedicated to training the classification prediction capability of the embedding space feature vectors based on labels and supervised training method.

E. Compression Ratio Estimation by KDE-SI

In hyperspectral applications under CS, high accuracy can often be achieved without full bands. Dispelling redundant bands helps reduce computation and storage burden. In the proposed HTD method, the anchor of the triple is independently sampled from the RDP prior distribution during training. We hope that the transformer can learn the dependencies between the spectra and the RDP prior distribution. However, when the compression ratio (CR) is reduced, the compressed spectra y may not accurately exhibit their Gaussian distribution, which can hinder the training process.

Therefore, the estimation of the CR can be expressed as determining the minimum number of bands that the compressed spectra exhibit Gaussian distribution. This is a problem of probability estimating, and KDE is a viable method. By using KDE-SI, we can identify the minimum number of bands M that satisfies the requirement.

KDE can be expressed by (24), where M_i represents the number of CS bands at the ith iteration, h is the bandwidth for smoothing, K is the Gaussian kernel function, y_t is the spectral vector under CS, and $\bar{f}_i(y)$ is the probability density function estimate for this number of bands. Using this formula allows us to estimate the probability density function of the current spectral vector at each iteration

$$\bar{f}_i(y) = \frac{1}{M_i h}\sum_{j=1}^{M_i} K\left(\frac{y - y_t(j)}{h}\right). \tag{24}$$

According to (7), the actual probability function $f_i(y)$ for M_i bands can be represented as follows:

$$f_i(y) = \frac{1}{\sqrt{2\pi B_L^2}}\exp\left(-\frac{y^2}{2B_L^2}\right). \tag{25}$$

Let $\xi = \sum_{l=1}^{L}\left(\mu^2(l) + \sigma^2(l)\right)$, and B_L^2 can be expressed as follows:

$$\begin{aligned} B_L^2 &= \frac{1}{M_i}\sum_{l=1}^{L}\left(\mu^2(l) + \sigma^2(l)\right) \\ &= \frac{1}{M_i}\xi. \end{aligned} \tag{26}$$

Then, the probability function can be expressed as follows:

$$f_i(y) = \frac{\sqrt{M_i}}{\sqrt{2\pi\xi}}\exp\left(-\frac{M_i y^2}{2\xi}\right). \tag{27}$$

The relationship between the probability density functions of the ith and $(i-1)$th iterations is described as follows:

$$\frac{f_i(y)}{f_{i-1}(y)} = \sqrt{\frac{M_i}{M_{i-1}}}\exp\left(-\frac{M_i y^2}{2\xi} + \frac{M_{i-1} y^2}{2\xi}\right)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

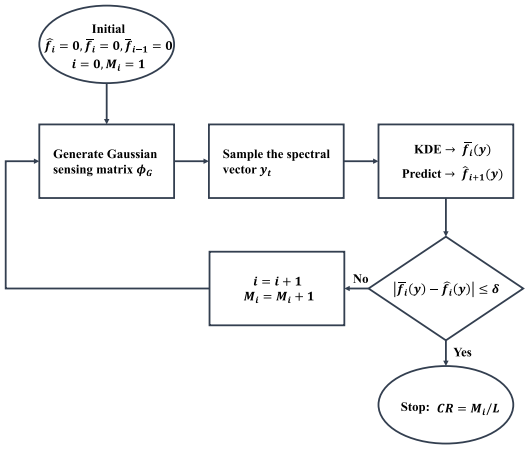


Fig. 6. Illustration of KDE-SI.

$$= \sqrt{\frac{M_i}{M_{i-1}}} \exp\left(\frac{(-M_i + M_{i-1})y^2}{2\xi}\right). \quad (28)$$

Assume an incremental step of 1 in iterations, $M_i - M_{i-1} = 1$. Since ξ is the sum of the squared means and variances of all bands y, it can be considered that $\xi \gg y^2$; then, the above equation can be further simplified to

$$\frac{f_i(y)}{f_{i-1}(y)} = \sqrt{\frac{M_i}{M_{i-1}}} = \sqrt{\frac{M_i}{M_i - 1}}. \quad (29)$$

According to (29), we can use the probability function estimation $\bar{f}_{i-1}(y)$ from the $(i-1)$th iteration to predict $\hat{f}_i(y)$ for the ith iteration, i.e.,

$$\hat{f}_i(y) = \overline{f}_{i-1} \sqrt{\frac{M_i}{M_{i-1}}}. \quad (30)$$

As the number of bands M_i increases and information acquisition improves, the prediction of $\hat{f}_i(y)$ becomes more accurate. Therefore, a discriminant formula can be constructed as in (2.7), setting a hard threshold δ as the stopping criterion

$$\left|\bar{f}_i(y) - \hat{f}_i(y)\right| = \left|\bar{f}_i(y) - \bar{f}_{i-1}(y)\sqrt{\frac{M_i}{M_{i-1}}}\right| \le \delta. \quad (31)$$

The flow of KDE-SI is demonstrated in Fig. 6. In the process, KDE step is achieved by (24), and predict step by (30).

IV. Experiment

A. Dataset

1) *San Diego Dataset:* The first dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the San Diego airport area in CA, USA. The images used in this experiment were not preprocessed to remove noise. The scene samples and ground truth are shown in the first and second columns of Fig. 5, respectively. Detailed information is provided in Table I.

2) *Airport–Beach–Urban (ABU) Dataset:* Another three scenes of the ABU dataset [40] were used and can be obtained from a public website. The first and second columns of Fig. 5 show scene samples and ground truth, respectively. The selected scenes include the airport, beach, and urban areas, all captured by the AVIRIS sensor. Detailed information is provided in Table I.

TABLE I
Details of the Datasets

	Sensor	resolution	Size	Original bands
Sandiego	AVIRIS	3.5m	100*100	184
ABU-Airport	AVIRIS	7.1m	100*100	205
ABU-Beach	AVIRIS	4.4m	150*150	188
ABU-Urban	AVIRIS	17.2m	100*100	204

A Gaussian sensing matrix was constructed based on (2) to perform CS on all datasets, generating a series of compressed hyperspectral images for detection. To construct prior information for the targets, we extracted all target spectra from the ground truth and used them as samples for the spectral random vector. Mean and variance were calculated for each discrete point, and the target distribution under CS was constructed based on (7) and (8).

B. Evaluation Criteria

1) ROC and AUC: The evaluation criteria include receiver operating characteristic (ROC) analysis [34] and the area under the curve (AUC). The ROC is composed of the detection rate (P_d), false alarm rate (P_f), and threshold (τ). Three curves, namely, ROC(P_d, P_f), ROC(P_d, τ), and ROC(P_f, τ), are formed by combining these components. AUC represents the area enclosed by the ROC curve and provides a more accurate reflection of the detector's performance compared with the ROC curve. The AUC(P_d, P_f) curve serves as an indicator of the trade-off between detection rate and false alarm rate, with a higher AUC indicating better detection performance. The AUC(P_f, τ) curve is designed to assess the background suppression ability of the detector, where a higher or lower AUC reflects better background suppression capability.

In addition, composite metric of three AUCs for all-around estimation is added, which can be expressed as follows:

$$\mathrm{AUC_{OA}} = \mathrm{AUC}(P_d, P_f) + \mathrm{AUC}(P_d, \tau) - \mathrm{AUC}(P_f, \tau) \quad (32)$$

$$\mathrm{AUC_{SPNR}} = \frac{\mathrm{AUC}(P_d, \tau)}{\mathrm{AUC}(P_f, \tau)}. \quad (33)$$

2) Separability Map: The boxplot is employed to analyze the separability between background and target [33], [49]. It accurately depicts the distribution of discrete data and is not influenced by outliers. The box encompasses the upper and lower quartiles of the data, with the center line representing the median, and the upper and lower boundaries indicating the limits after excluding outliers. The distance between the boxes of background and target provides a visual reflection of their separability. If there is an overlap between the background and target boxes, the detection results may lead to misjudgments. The size of the box reflects the degree of aggregation within the same category.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

C. Comparison Methods

In this experiment, we compared the detection performance of the proposed algorithm in original hyperspectral images with its performance in different CS bands. The selected benchmark algorithms for comparison include OSP, ACE, CEM, ECEM,[1] SMF, HTD-IRN[2] [50], MLSN,[3] CNND[4] [51], and SSDCTD[5] [52].

For the proposed algorithm in this article, we conduct comparative experiments with different CRs. Let the original number of spectral bands in the dataset be L, and the number of CS bands be M. We define CR as M/L, where a smaller ratio indicates a higher level of compression for the hyperspectral image. When constructing the network, the transformer has two layers, each layer had two heads in the MultiHead attention, and the FNN dimension is set to 64. In addition, CCN includes two CombineCov layers. During training, both stages have 400 iterations. In the first stage, the learning rate (lr) is set to $1e-4$, and weight decay to $1e-2$. In the second stage, lr is set to $5e-4$, the margin alpha for triplet loss is 1, and β for semi-HNM in the second stage is set to 0.1.

D. Detection Results

1) Detection Map Comparison: Figs. 7–10 illustrate the grayscale detection results: (a) and (b) are the original hyperspectral pseudocolor image and the ground-truth map. (c)–(e) show the detection results of the proposed method at CR $= [0.25, 0.50, 0.75]$, while (f)–(m) display the results of the comparative algorithms.

The proposed algorithm demonstrates superior target detection performance across four datasets. It consistently maintains high clarity and completeness of target contours, although it shows some background interference in certain areas. Compared with other methods, it effectively extracts target pixels and preserves detailed contours. While ACE, ECEM, and SMF exhibit strong background suppression, they often compromise target integrity, resulting in blurred or incomplete detection. HTD-IRN and MLSN generally perform poorly, struggling to distinguish targets from the background and often failing to maintain target integrity.

Overall, the proposed algorithm provides clearer and more accurate detection results, outperforming other methods in maintaining target detail and minimizing background noise.

2) Separability Map Comparison: Separability map of the detection results for various algorithms on four datasets is plotted in Fig. 11. The box boundaries represent the upper and lower quartiles, the center denotes the median, and the whiskers indicate the upper and lower bounds. The right curve represents a fitted normal distribution.

On the San Diego dataset, the proposed algorithm exhibits significantly higher separability, with background and target box boundaries almost merging, closely approaching 1 and 0. For the ABU-Airport dataset, the proposed algorithm's background predictions cluster around 0, while target pixel scores concentrate around 1 for larger CRs. In comparison, other algorithms show broader intraclass distribution ranges for targets or closer proximity to background predictions.

Concerning the ABU-Beach dataset, the proposed algorithm demonstrates superior target score concentration for CR greater than 0.50, outperforming other algorithms. Although the background distribution is less concentrated than in the previous dataset, the separability is still superior to other algorithms. On the ABU-Urban dataset, similar to ABU-Beach, the target prediction scores are more concentrated for larger CR, and the background box closely aligns with 0.

In summary, in the separability maps, the prediction scores of the proposed algorithm for target and background are aggregated around 1 and 0, respectively. This leads to significantly better separability between targets and background compared with comparative algorithms.

3) ROC and AUC Comparison: Fig. 12 shows the ROC curves for the detection results of various algorithms on the four datasets. Moving from left to right, the curves represent ROC(P_d, P_f), ROC(P_f, τ), and ROC(P_d, τ). Observing the curve shapes, the proposed algorithm's ROC(P_d, P_f) closely aligns with the top-left corner on all datasets, and ROC(P_d, τ) adheres well to the top-right corner. On the ABU-Beach dataset, ACE and MLSN have ROC(P_f, τ) closer to the bottom-left corner, while the proposed algorithm is closer to ECEM.

Detailed description of the AUCs is shown in Tables II–V, with processing times. Except for the HTD processing time, the proposed algorithm achieves majority of best performance (bold) on the four datasets.

Regarding single HTD processing time, the proposed algorithm takes approximately 0.04 s on the San Diego, ABU-Airport, and ABU-Urban datasets, and around 0.07 s on the ABU-Beach dataset. As the CR increases, leading to more bands, processing time will increase. SSDCTD has the shortest processing time on all datasets, followed by the proposed method.

Overall, the proposed method achieves the best results on almost all the metrics, and the processing time is significantly lower than other DL methods.

4) Model Size Comparison: We compare the total size of models in float32 format for the proposed network with other HTD algorithm based on DL in Table VI, specifically HTD-IRN and MLSN. The results are presented in the table below. The proposed network, which configures with two transformer layers, two heads in the MultiHead attention mechanism, and two layers of CombineCov layer, has significantly fewer parameters compared with the counterparts.

5) Ablation Experiment: In the ablation experiment, we first explore the effectiveness of the CCN module. Using traditional metrics, namely, Euclidean distance and cosine similarity, we measure the similarity of transformer embedding feature vectors obtained after completing the first training stage. The similarity serves as the prediction result. Contrasting with the prediction results from the embedding space using the CCN module, trained in the second stage, is shown in Table VII. It can be observed that the proposed CCN module surpasses

[1]https://github.com/ruizhaocv/E_CEM-for-Hyperspectral-Target-Detection
[2]https://github.com/shendb2022/HTD-IRN
[3]https://github.com/YuleiWang1/MLSN
[4]https://github.com/yuchaoY/CNND
[5]https://github.com/zxd52csx/Pixel-level-Hyperspectral-Target-Detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

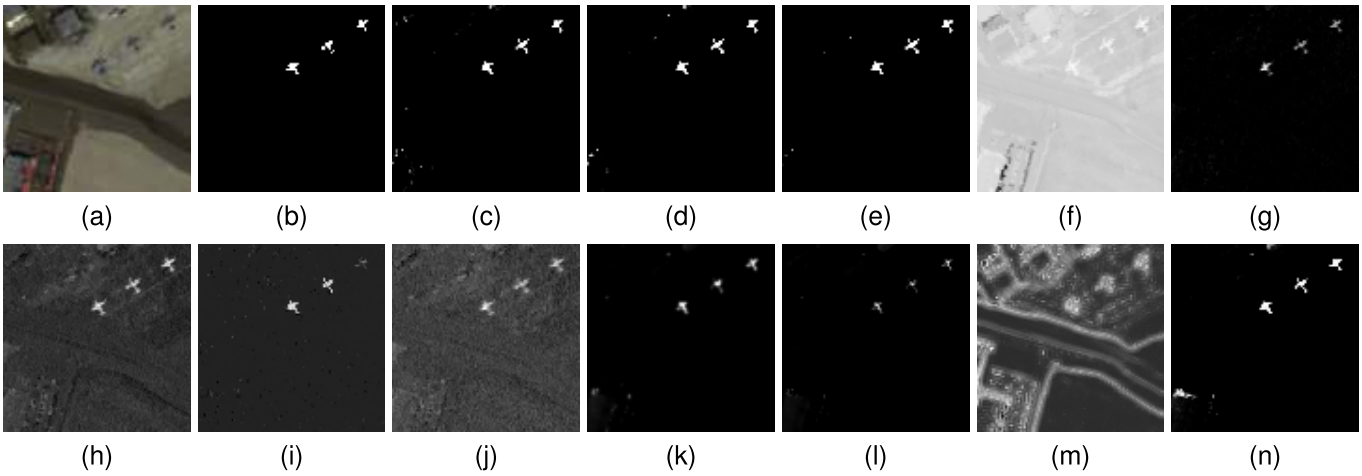


Fig. 7. Detection results of the San Diego dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.25. (d) Ours CR = 0.5. (e) Ours CR = 0.75. (f) OSP. (g) ACE. (h) CEM. (i) ECEM. (j) SMF. (k) HTD-IRN. (l) MLSN. (m) CNND. (n) SSDCTD.

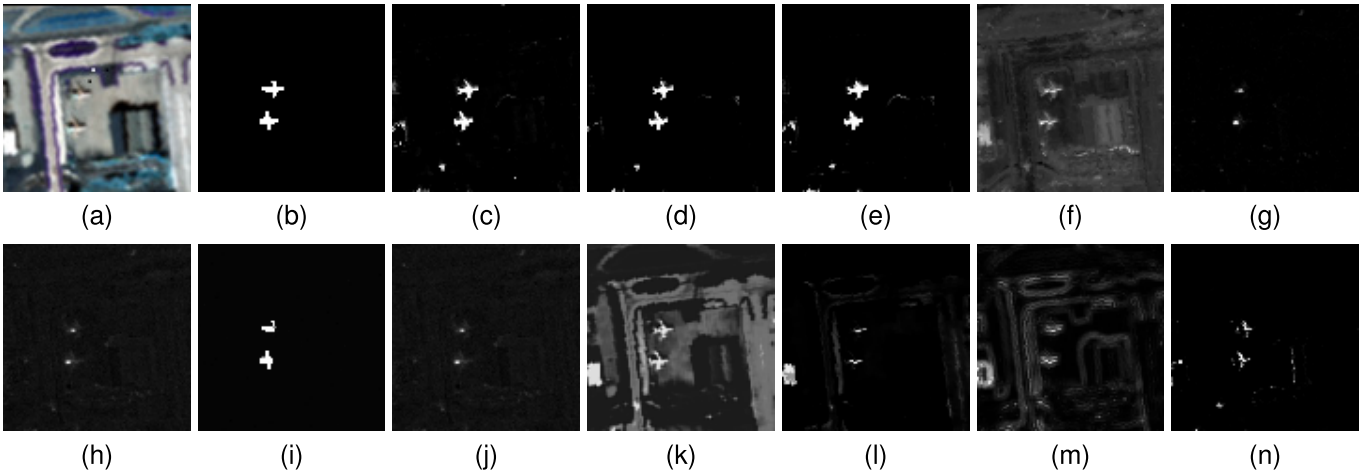


Fig. 8. Detection results of the ABU-Airport dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.25. (d) Ours CR = 0.5. (e) Ours CR = 0.75. (f) OSP. (g) ACE. (h) CEM. (i) ECEM. (j) SMF. (k) HTD-IRN. (l) MLSN. (m) CNND. (n) SSDCTD.

traditional metrics in all metrics. In particular, in terms of AUC(P_f, τ), the false positive rate of the CCN module's detection results is reduced by two orders of magnitude compared with traditional metrics, significantly enhancing background suppression capability and signal-to-noise ratio.

Next, we investigate the impact of the two-stage training approach on the network's detection performance. In the comparison, we set the training parameters for the one-stage approach to be consistent with the two-stage approach, with the following loss functions. Unlike two-stage training, triplet loss and BCE loss are used simultaneously to update both transformer and CCN

$$L = L_{\text{Triplet}} + L_{\text{BCE}}. \tag{34}$$

The comparative results are shown in Table VIII, indicating a sharp decline in performance with the one-stage training approach. Especially, at higher CRs, there are cases where the AUC_{OA} becomes negative, and the area under the false positive rate curve surpasses the area under the detection rate curve.

In conclusion, the ablation experiments show that the proposed CCN and training approach in this article can significantly improve target detection performance.

E. CR Estimation

In this experiment, the proposed KDE-SI is performed to estimate the spectral probability density function of the San Diego dataset. The trend of average $\left|\bar{f}_i(y) - \hat{f}_i(y)\right|$ is shown on the left vertical axis in Fig. 13. This curve indicates that as the number of CS bands increases, the probability distribution of the spectral vectors tends to stabilize.

The right vertical axis of Fig. 13 shows the variation of AUC_{OA} with M. It is evident that the turning points of both the AUC_{OA} curve and the KDE-SI occur around $M = 30$, with CR ≈ 0.16. This indicates that the proposed method can achieve stable detection performance when CR > 0.16.

F. Discussion

The experiments on public datasets, comparing our algorithm with the existing state-of-the-art ones in the original domain, demonstrate the excellence of our proposed HTD algorithm in CS.

1) Analysis of the detection map reveals that the preservation of target integrity is one of the advantages of the proposed algorithm. There is minimal loss of target pixels across all four datasets. In particular, in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AUC VALUES AND HTD TIME ON THE SAN DIEGO DATASET

	AUC(P_d, P_f)	**AUC**(P_f, τ)	**AUC**(P_d, τ)	**AUC$_{OA}$**	**AUC$_{SPNR}$**	**Times**$/s$
Ours CR = 0.25	**0.999981**	0.002232	0.990463	1.988212	443.752927	0.035856
Ours CR = 0.50	0.999948	0.002219	0.992725	1.990454	447.359150	0.038786
Ours CR = 0.75	0.999930	**0.001212**	**0.999079**	**1.997798**	**824.554800**	0.040267
OSP	0.992449	0.806598	0.934306	1.120157	1.158329	0.442430
ACE	0.999686	0.005954	0.469767	1.463499	78.895252	0.090702
CEM	0.999625	0.198381	0.694461	1.495706	3.500649	0.039828
ECEM	0.980330	0.133470	0.609380	1.456240	4.565670	4.319890
SMF	0.999159	0.255327	0.651214	1.395047	2.550514	0.042235
HTD-IRN	0.998700	0.003600	0.532800	1.527900	148.000000	0.196960
MLSN	0.996625	0.002017	0.338493	1.333101	167.835810	13.486038
CNND	0.959507	0.310210	0.642605	1.291902	2.071516	11.385513
SSDCTD	0.998174	0.004827	0.867508	1.860855	179.714534	**0.032985**

TABLE III
AUC VALUES AND HTD TIME ON THE ABU-AIRPORT DATASET

	AUC(P_d, P_f)	**AUC**(P_f, τ)	**AUC**(P_d, τ)	**AUC$_{OA}$**	**AUC$_{SPNR}$**	**Times**$/s$
Ours CR = 0.25	0.997814	0.008926	0.823250	1.812138	92.230896	0.038101
Ours CR = 0.50	0.998488	**0.003035**	0.803531	1.798983	**264.724175**	0.041223
Ours CR = 0.75	**0.998711**	0.005707	**0.874887**	**1.867891**	153.300435	0.042420
OSP	0.957747	0.193554	0.451937	1.216129	2.334932	0.437300
ACE	0.850747	0.003711	0.126362	0.973398	34.050230	0.098120
CEM	0.888639	0.067928	0.186728	1.007439	2.748925	0.043599
ECEM	0.926590	0.022810	0.693150	1.596930	30.387988	5.778010
SMF	0.888639	0.067928	0.186728	1.007439	2.748925	0.041762
HTD-IRN	0.831800	0.221300	0.594300	1.204800	2.685495	0.220933
MLSN	0.755158	0.017732	0.161612	0.899037	9.113909	14.246319
CNND	0.926569	0.068407	0.310654	1.168816	4.541285	11.446353
SSDCTD	0.947293	0.003696	0.243320	1.186917	65.828175	**0.020043**

TABLE IV
AUC VALUES AND HTD TIME ON THE ABU-BEACH DATASET

	AUC(P_d, P_f)	**AUC**(P_f, τ)	**AUC**(P_d, τ)	**AUC$_{OA}$**	**AUC$_{SPNR}$**	**Times**$/s$
Ours CR = 0.25	0.993447	0.026627	0.829063	1.795883	31.135794	0.069548
Ours CR = 0.50	**0.997966**	0.038644	**0.948523**	**1.907845**	24.545022	0.072314
Ours CR = 0.75	0.995464	0.060183	0.937110	1.872391	15.571049	0.073239
OSP	0.832902	0.368712	0.447201	0.911392	1.212874	0.659995
ACE	0.836491	**0.003435**	0.222250	1.055307	**64.705870**	0.192510
CEM	0.898670	0.156497	0.391350	1.133523	2.500685	0.062187
ECEM	0.779020	0.021843	0.163470	0.920647	7.483862	5.822886
SMF	0.898670	0.156497	0.391350	1.133523	2.500685	0.101260
HTD-IRN	0.930910	0.251582	0.542098	1.221426	2.154757	0.316391
MLSN	0.971269	0.026915	0.553333	1.497687	20.558341	28.422527
CNND	0.835867	0.263975	0.369201	0.941093	1.398619	24.189681
SSDCTD	0.983188	0.015174	0.448184	1.416198	29.535632	**0.040073**

ABU-Airport dataset, the detection maps of other algorithms depict airplanes almost degraded to point regions, while the proposed algorithm effectively preserves the contours of the targets. In addition, from the ROC(P_d, τ) curves in Fig. 12 and the AUC(P_d, τ) values in the table, it is evident that the proposed method achieves extremely high target detection rates, with the AUC almost approaching to 1. This is attributed to the fact that other algorithms rely on a single spectrum as the basis for matching targets and cannot adapt well to variations in target spectra. Even spectra belonging to positive samples (targets) exhibit different spectral characteristics in different regions. The proposed method, through RDP, learns the distribution of spectral random vectors caused by spectral variations under CS. Furthermore, during training, continuous resampling of this distribution ensures a more comprehensive set of spectral samples for the target class. Experimental results demonstrate the effectiveness of this strategy.

2) Further analysis of the separability map indicates that the proposed method exhibits strong background suppression capabilities across all four datasets. The transformer increases the distance between positive and negative samples in the embedding space, and the subsequent CCN further amplifies the differences in their classification predictions. This strategy is particularly effective in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

TABLE V
AUC VALUES AND HTD TIME ON THE ABU-URBAN DATASET

	$\mathbf{AUC}(P_d, P_f)$	$\mathbf{AUC}(P_f, \tau)$	$\mathbf{AUC}(P_d, \tau)$	$\mathbf{AUC_{OA}}$	$\mathbf{AUC_{SPNR}}$	$\mathbf{Times}/s$
Ours CR = 0.25	**0.999973**	0.007334	0.917845	1.910484	125.151301	0.034052
Ours CR = 0.50	0.999873	0.008464	**0.994367**	**1.985777**	117.485440	0.038878
Ours CR = 0.75	0.999936	**0.001413**	0.977873	1.976397	**692.240460**	0.041980
OSP	0.923473	0.304209	0.641051	1.260315	2.107269	0.367290
ACE	0.966237	0.004189	0.236155	1.198203	56.371534	0.093668
CEM	0.983532	0.088669	0.317077	1.211940	3.575956	0.041175
ECEM	0.923250	0.009070	0.682250	1.596430	75.220507	5.938330
SMF	0.983532	0.088669	0.317077	1.211940	3.575956	0.042787
HTD-IRN	0.979300	0.009800	0.200200	1.169700	20.428571	0.225750
MLSN	0.984246	0.010710	0.429287	1.402823	40.083788	17.048152
CNND	0.990863	0.066964	0.513621	1.437520	7.670150	11.166008
SSDCTD	0.996815	0.002073	0.770066	1.764808	371.412850	**0.020007**

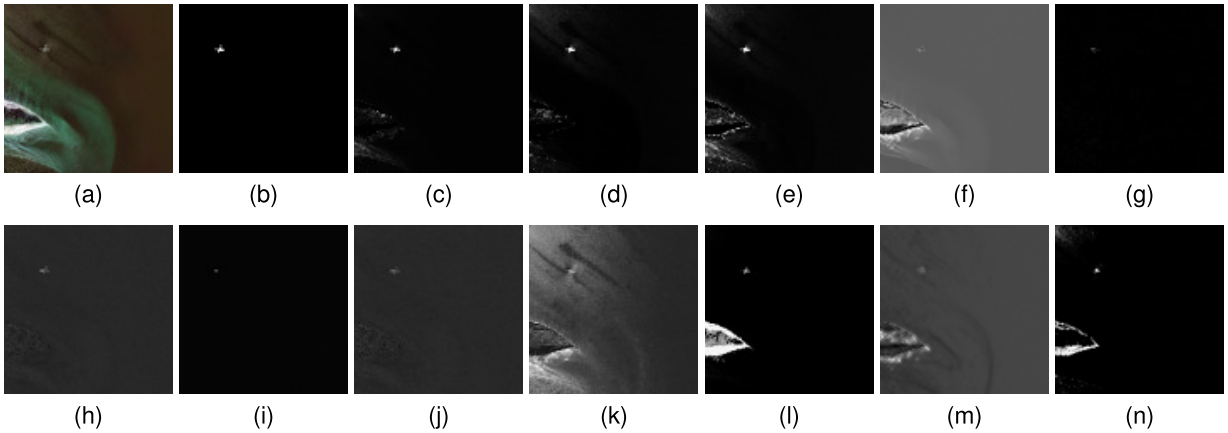


Fig. 9. Detection results of the ABU-Beach dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.25. (d) Ours CR = 0.5. (e) Ours CR = 0.75. (f) OSP. (g) ACE. (h) CEM. (i) ECEM. (j) SMF. (k) HTD-IRN. (l) MLSN. (m) CNND. (n) SSDCTD.

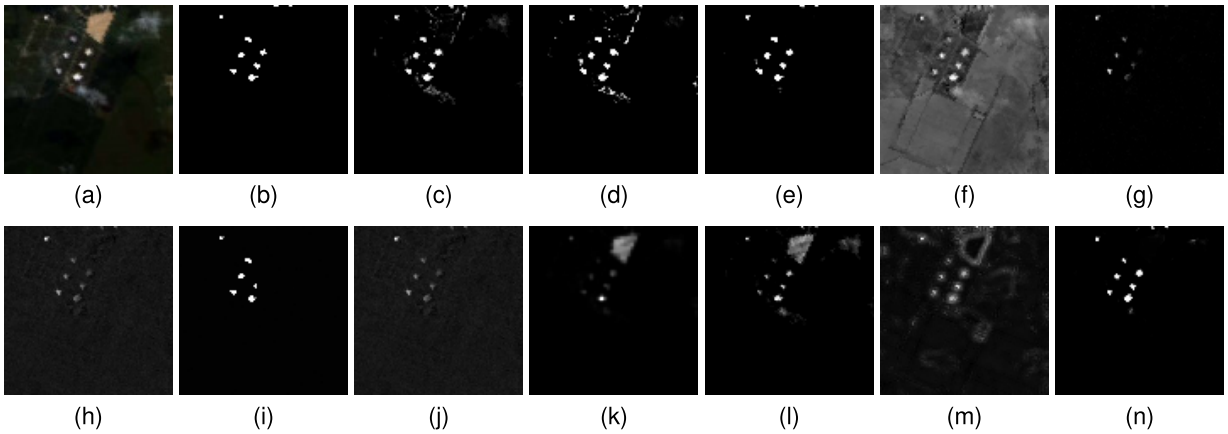


Fig. 10. Detection results of the ABU-Urban dataset. (a) Original image. (b) Ground truth. (c) Ours CR = 0.25. (d) Ours CR = 0.5. (e) Ours CR = 0.75. (f) OSP. (g) ACE. (h) CEM. (i) ECEM. (j) SMF. (k) HTD-IRN. (l) MLSN. (m) CNND. (n) SSDCTD.

classifying negative samples, as seen in Fig. 11, where almost all background categories aggregate near 0. In terms of ROC, this approach demonstrates lower AUC(P_f, τ), as shown in Tables II–V, achieving curve areas on the order of 10^{-3} for all datasets except ABU-Beach.

3) Analyzing the ABU-Beach dataset, which exhibits poor background suppression performance, reveals significant background interference in the bottom-left portion of the detection map, as shown in Fig. 9. In addition, the AUC(P_f, τ) is lower than that of ACE and ECEM in

TABLE VI
COMPARISON OF MODEL SIZE

Methods	Model Size/kb
HTD-IRN	1646.59
MLSN	202.78
Ours	98.84

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

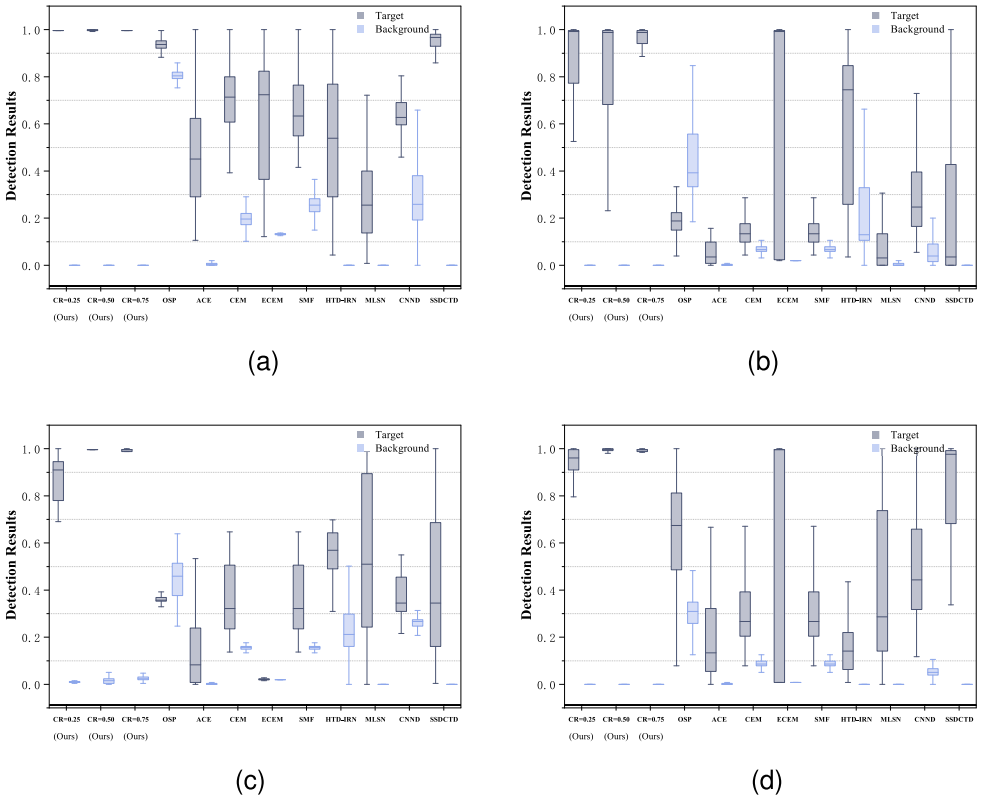


Fig. 11. Separability map of detection results. (a) San Diego. (b) ABU-Airport. (c) ABU-Beach. (d) ABU-Urban.

TABLE VII
COMPARISON CCN WITH TRADITIONAL METRICS

CR	Method	$AUC(P_d, P_f)$	$AUC(P_f, \tau)$	$AUC(P_d, \tau)$	$AUC_{\mathbf{OA}}$
Ours CR = 0.25	**Transformer+Euclidean distance**	0.999974	0.256717	0.675856	1.419113
	Transformer+CosSimilarity	0.999818	0.342797	0.738261	1.395282
	Transformer+CCN	**0.999984**	**0.002198**	**0.990497**	**1.988283**
Ours CR = 0.50	**Transformer+Euclidean distance**	0.999632	0.418285	0.773523	1.35487
	Transformer+CosSimilarity	0.999823	0.441818	0.781455	1.33946
	Transformer+CCN	**0.999951**	**0.002216**	**0.992728**	**1.990463**
Ours CR = 0.75	**Transformer+Euclidean distance**	0.999971	0.396669	0.752235	1.355537
	Transformer+CosSimilarity	0.999414	0.468669	0.828560	1.359305
	Transformer+CCN	**0.999988**	**0.001154**	**0.999137**	**1.997971**

TABLE VIII
COMPARISON OF TRAINING STYLE

CR	Training Method	$AUC(P_d, P_f)$	$AUC(P_f, \tau)$	$AUC(P_d, \tau)$	$AUC_{\mathbf{OA}}$
Ours CR = 0.25	**One-stage**	0.994069	0.236869	0.587318	1.344518
	Two-stage	**0.999984**	**0.002198**	**0.990497**	**1.988283**
Ours CR = 0.50	**One-stage**	0.132048	0.809004	0.739165	0.062209
	Two-stage	**0.999951**	**0.002216**	**0.992728**	**1.990463**
Ours CR = 0.75	**One-stage**	0.057688	0.765138	0.540454	-0.166996
	Two-stage	**0.999988**	**0.001154**	**0.999137**	**1.997971**

Table IV. However, in the separability map [Fig. 11(c)], there is no indication of nonaggregation of positive sample detection results, suggesting that the network is overly confident in its classification outcomes. This is attributed to insufficient prior information on this dataset. There are only 19 spectral vectors serving as target priors for ABU-Beach, while other datasets have 58, 87, and 67 target priors, respectively. This leads to inaccurate estimates of μ_t and σ_t^2 for targets in the original domain, causing the distribution calculated through RDP to inaccurately cover CS target spectral vectors.

4) The ablation experiments indicate that, at least in the case of a small model size, CCN is more effective

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

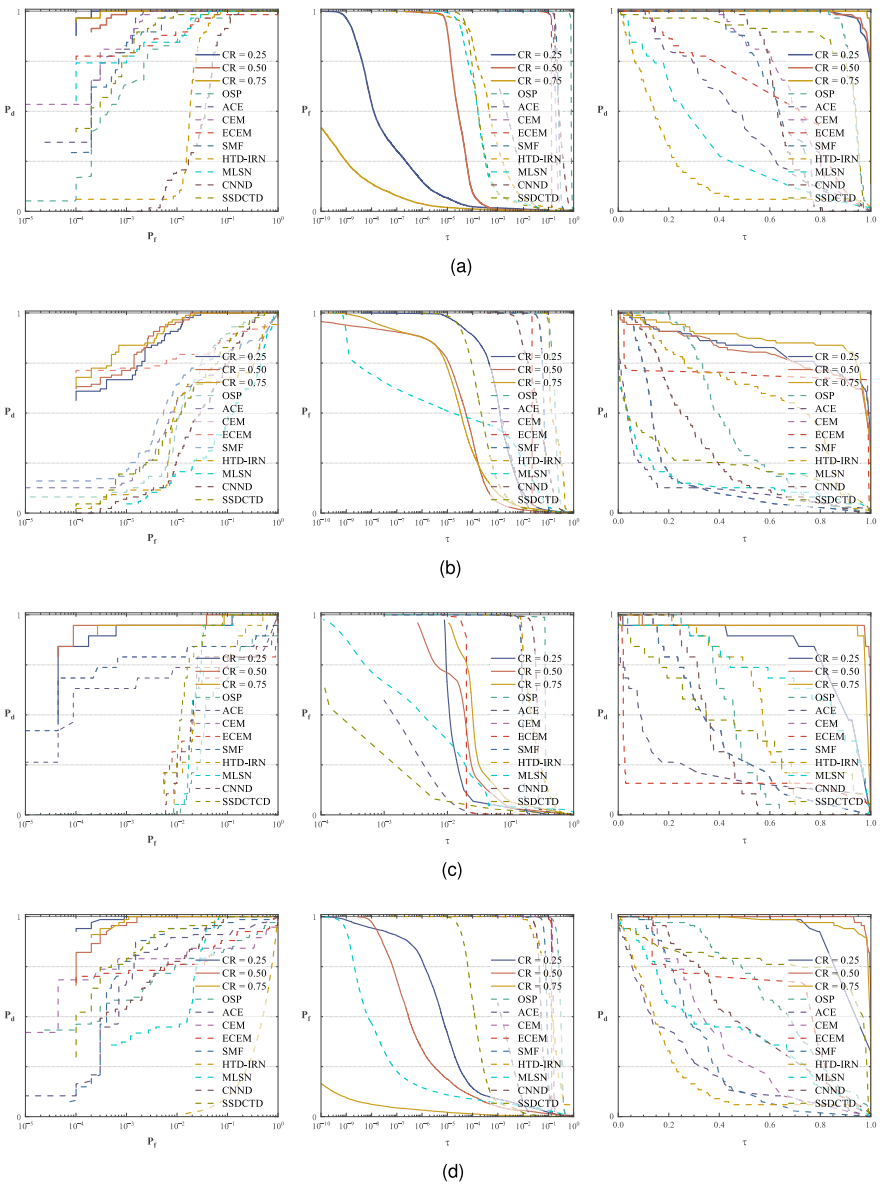


Fig. 12. ROC curves of detection results. (a) San Diego. (b) ABU-Airport. (c) ABU-Beach. (d) ABU-Urban.

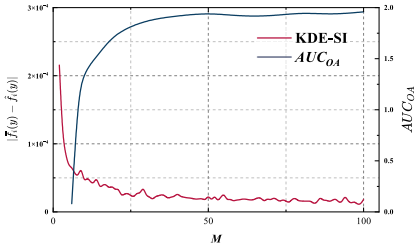


Fig. 13. (Left) Iteration value in KDE-SI. (Right) Changing of AUC_{OA} with M.

than traditional similarity measurement methods. Our network, with a smaller size compared with other networks of the same type, faces the challenge of learning an embedding mapping that can completely distinguish between positive and negative samples with limited parameters. It is difficult to achieve this solely by stacking transformer structures. CCN complements the transformer's limited ability to capture local information by combining convolutions in the embedding space, ultimately achieving high detection performance with reduced network parameters. This is particularly significant for resource-constrained HTD.

5) The proposed method achieves relatively stable detection performance in different CRs. In Fig. 13, the minimum CR required to maintain high performance is estimated. The results indicate that for the San Diego dataset, a CR of approximately 0.16 is the threshold for maintaining this detection performance. This means that the proposed method can save the computation and storage consumption by reducing the number of compressed bands without significantly compromising detection performance and takes the full advantage of compressed HTD.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In conclusion, our study introduces a novel CS-TTD approach for achieving HTD without CS reconstruction. The CS-TTD involves data augmentation via RDP, followed by a triplet transformer as the backbone and CCN for classification.

The RDP-based augmentation expands the number of targets by sampling the target distribution in CS, which is also conducive to the triplet transformer effectively capturing global features with long-range dependencies. The CCN is an alternative to traditional similarity metrics for classification, capturing the combined local features of spectral embedding vectors. The joint training is achieved by an improved two-stage ISIA training approach, incorporating triplet loss and labeled BCE loss.

The KDE-SI suggests that CS-TTD can maintain performance while reducing CR to approximately 0.16. Comparative experiments, including ablation studies on four datasets, also demonstrate the effectiveness of our proposed method in the compressed domain.

References

[1] Y. Zhang et al., "Inversion study of heavy metals in soils of potentially polluted sites based on UAV hyperspectral data and machine learning algorithms," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Mar. 2021, pp. 1–5.

[2] W. Ma, K. Tan, Q. Du, J. Ding, and Q. Yan, "Estimating soil heavy metal concentration using hyperspectral data and weighted K-NN method," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–4.

[3] N. Kobayashi, A. Yoshino, M. Ishikawa, and S. Homma, "Anemia examination using a hyperspectral camera in telecare system," in *Proc. IEEE 3rd Global Conf. Life Sci. Technol. (LifeTech)*, Mar. 2021, pp. 475–476.

[4] A. Lugovtsov et al., "Erythrocyte size distribution retrieval via laser diffractometry and hyperspectral holography of blood smears," in *Proc. Int. Conf. Laser Opt. (ICLO)*, Jun. 2018, p. 511.

[5] L. Antson, A. Vandenhoeke, M. Shimoni, C. Hamesse, and H. Luong, "Detection and tracking of search and rescue personnel under hindered light conditions using hyperspectral imaging," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–6.

[6] Y. Du, J. Jiang, Z. Liu, and Y. Pan, "Combining a crop growth model with CNN for underground natural gas leakage detection using hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1846–1856, 2022.

[7] F. Tanriverdi, D. Schuldt, and J. Thiem, "Hyperspectral imaging: Color reconstruction based on medical data," in *Proc. IEEE-EMBS Conf. Biomed. Eng. Sci. (IECBES)*, Dec. 2018, pp. 194–199.

[8] G. Zhan, Y. Uwamoto, and Y.-W. Chen, "HyperUNet for medical hyperspectral image segmentation on a choledochal database," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2022, pp. 1–5.

[9] K. Chen, "Military object detection using multiple information extracted from hyperspectral imagery," in *Proc. Int. Conf. Prog. Inform. Comput. (PIC)*, Dec. 2017, pp. 124–128.

[10] V. Křivánek, J. Motsch, and Y. Bergeon, "Hyperspectral data acquisition for military camouflage in vegetation—Preliminary results," in *Proc. Commun. Inf. Technol. (KIT)*, Oct. 2023, pp. 1–7.

[11] Y. Montembeault, P. Lagueux, V. Farley, A. Villemaire, and K. C. Gross, "Hyper-cam: Hyperspectral IR imaging applications in defence innovative research," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[12] J. Zhao, G. Wang, B. Zhou, J. Ying, and J. Liu, "SRA–CEM: An improved CEM target detection algorithm for hyperspectral images based on subregion analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6026–6037, 2023.

[13] L. Sun, Z. Ma, and Y. Zhang, "ABLAL: Adaptive background latent space adversarial learning algorithm for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 411–427, 2024.

[14] E. J. Candes and M. B. Wakin, "An introduction to compressive sampling," *IEEE Signal Process. Mag.*, vol. 25, no. 2, pp. 21–30, Mar. 2008.

[15] S. Zhang, H. Huang, and Y. Fu, "Fast parallel implementation of dual-camera compressive hyperspectral imaging system," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 11, pp. 3404–3414, Nov. 2019.

[16] C. Kwan et al., "Target tracking and classification using compressive measurements of MWIR and LWIR coded aperture cameras," *J. Signal Inf. Process.*, vol. 10, no. 3, pp. 73–95, 2019.

[17] C. Kwan et al., "Target tracking and classification using compressive sensing camera for SWIR videos," *Signal, Image Video Process.*, vol. 13, no. 8, pp. 1629–1637, Nov. 2019.

[18] C. Kwan, B. Chou, J. Yang, and T. Tran, "Deep learning based target tracking and classification for infrared videos using compressive measurements," *J. Signal Inf. Process.*, vol. 10, no. 4, pp. 167–199, 2019.

[19] Q. Du, H. Ren, and C.-I. Chang, "A comparative study for orthogonal subspace projection and constrained energy minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1525–1529, Jun. 2003.

[20] S. Kraut, L. L. Scharf, and L. T. McWhorter, "Adaptive subspace detectors," *IEEE Trans. Signal Process.*, vol. 49, no. 1, pp. 1–16, Jan. 2001.

[21] C.-I. Chang and D. C. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.

[22] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.

[23] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, Jun. 2019.

[24] Q. Ling, Y. Guo, Z. Lin, L. Liu, and W. An, "A constrained sparse-representation-based binary hypothesis model for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1933–1947, Jun. 2019.

[25] Y. Niu and B. Wang, "Hyperspectral target detection using learned dictionary," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1531–1535, Jul. 2015.

[26] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[27] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[28] S. Chopra, R. Hadsell, and Y. LeCun, "Learning a similarity metric discriminatively, with application to face verification," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2005, pp. 539–546.

[29] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Boston, MA, USA, Jun. 2015, pp. 4353–4361.

[30] J.-H. Kim, J. Kim, and J. Joung, "Siamese hyperspectral target detection using synthetic training data," *Electron. Lett.*, vol. 56, no. 21, pp. 1116–1118, Oct. 2020.

[31] Z. Dou, K. Gao, X. Zhang, J. Wang, and H. Wang, "Deep learning-based hyperspectral target detection without extra labeled data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 1759–1762.

[32] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[33] J. Jiao, Z. Gong, and P. Zhong, "Triplet spectralwise transformer network for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519817.

[34] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527913.

[35] Y. Shi, J. Li, Y. Yin, B. Xi, and Y. Li, "Hyperspectral target detection with macro-micro feature extracted by 3-D residual autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4907–4919, Dec. 2019.

[36] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.

[37] W. Xie, J. Zhang, J. Lei, Y. Li, and X. Jia, "Self-spectral learning with GAN based spectral–spatial target detection for hyperspectral image," *Neural Netw.*, vol. 142, pp. 375–387, Oct. 2021.
[38] W. Xie, X. Zhang, Y. Li, K. Wang, and Q. Du, "Background learning based on target suppression constraint for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5887–5897, 2020.
[39] H. Gao, Y. Zhang, Z. Chen, F. Xu, D. Hong, and B. Zhang, "Hyperspectral target detection via spectral aggregation and separation network with target band random mask," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515516.
[40] Q. Yang, X. Wang, D. Wang, B. Yu, Y. Zhou, and S. Qiao, "Compressive hyperspectral target detection with restricted distribution property," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517815.
[41] M. E. Lopes, "Estimating unknown sparsity in compressed sensing," 2012, *arXiv:1204.4227*.
[42] B. Lampe, C.-I. Chang, A. Bekit, and C. D. Porta, "Restricted entropy and spectrum properties for the compressively sensed domain in hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5642–5652, Aug. 2020.
[43] C. J. Della Porta and C.-I. Chang, "Estimating optimal number of compressively sensed bands for hyperspectral classification via feature selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11775–11788, 2021.
[44] C. J. Della Porta and C.-I. Chang, "Progressive compressively sensed band processing for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2378–2390, Mar. 2021.
[45] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[46] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 4685–4694.
[47] R. Baraniuk, M. Davenport, R. DeVore, and M. Wakin, "A simple proof of the restricted isometry property for random matrices," *Constructive Approximation*, vol. 28, no. 3, pp. 253–263, Dec. 2008.
[48] C. J. D. Porta, A. A. Bekit, B. H. Lampe, and C.-I. Chang, "Hyperspectral image classification via compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8290–8303, Oct. 2019.
[49] Y. Liu, W. Xie, Y. Li, Z. Li, and Q. Du, "Dual-frequency autoencoder for anomaly detection in transformed hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523613.
[50] D. Shen, X. Ma, W. Kong, J. Liu, J. Wang, and H. Wang, "Hyperspectral target detection based on interpretable representation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519416, doi: 10.1109/TGRS.2023.3302950.
[51] W. Li, G. Wu, and Q. Du, "Transferred deep learning for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5177–5180, doi: 10.1109/IGARSS.2017.8128168.
[52] X. Zhang et al., "Self-supervised learning with deep clustering for target detection in hyperspectral images with insufficient spectral variation prior," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103405.



Qingyao Yang received the B.S. degree from the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China, in 2022. He is currently pursuing the master's degree in electronic information engineering with the Institute of Microelectronics, Chinese Academy of Sciences, Beijing.

His research interests include remote sensing, image processing, and deep learning.



Xiaoqin Wang is currently a Researcher with the Institute of Microelectronics, Chinese Academy of Sciences, Beijing, China.

Her main research interests include low-power processors and microsystems design for intelligent perception, intelligent edge computing, and audio and video processing.



Long Chen received the B.S. degree from the Department of Precision Instrument, Tsinghua University, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree in electronic information engineering with the Institute of Microelectronics, Chinese Academy of Sciences, Beijing.

His research interests include artificial intelligence (AI) and hardware–software codesign of neural networks.



Yumei Zhou received the B.S. degree from the Department of Radio Electronics, Tsinghua University, Beijing, China, in 1985.

Since 1997, she has been a Researcher with the Institute of Microelectronics, Chinese Academy of Sciences, Beijing. Her research interests include integrated circuit design technology, reliability technology, low-power circuit design, and high-speed interface circuits.



Shushan Qiao (Member, IEEE) received the M.S. and Ph.D. degrees from the Institute of Microelectronics, Chinese Academy of Sciences, Beijing, China, in 2008.

Since 2008, he has been a Researcher with the Institute of Microelectronics, Chinese Academy of Sciences, where he has been the Deputy Director with the Smart Sensing Center, since 2017. His research interests include artificial intelligence, ultralow-power processors, intelligent microsystems, communication chips, and low-power integrated circuit design methodology.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:02:48 UTC from IEEE Xplore. Restrictions apply.